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MILITARY SPECIFICATION

ADAPTER, TEST MX-3413/GWM-4

1. SCOPE. This specification establishes the requirements for one type of electrical adapter designated as MX-3413/GWM-4 hereafter referred to as the adapter.

2. APPLICABLE DOCUMENTS. The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4158	Electronic Equipment Ground: General Requirements For
MIL-T-4807	Tests: Vibration and Shock Ground Electronic Equipment (Requirements For)
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification For
MIL-T-9107	Test Reports, Preparation Of
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery Of
MIL-C-25200	Cable Assemblies, Special Weapons, Electrical, General Requirements For
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-16	Electric and Electronic Reference Designations
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1190

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DRAWINGS

Air Force

60D46020	Adapter, Test MX-3413/GWM-4
60D46021	Schematic Diagram

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION. This specification makes provisions for preproduction testing.

3.2 GENERAL SPECIFICATIONS. The requirements of MIL-E-4158 and MIL-C-25200 are applicable as requirements of this specification. When conflict occurs between this specification and the general specification, the requirements of this specification shall govern.

3.3 ANCILLARY EQUIPMENT. The following items are used with, but not part of the adapters:

	Nomenclature	Quantity
a.	Cable Assembly, Special Purpose, Electrical CX-6683/U (used to test the F-100 and F-105/GAM-83B)	1
b.	Cable Assembly, Special Purpose, Electrical, Branched CX-6682/GWM-4	1
c.	Cable Assembly, Special Purpose, Electrical CX-6474/U (used to test the F-100 and F-105/GAM-83A)	1

3.4 DESIGN AND CONSTRUCTION. The design and construction of the adapter shall be in conformance to Air Force drawing 60D46020, subsidiary drawings and specifications and standards referenced therein.

3.4.1 CABLES. Design and construction of the cables shall be in accordance with MIL-C-25200.

3.4.2 INTERCHANGEABILITY. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When

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parts are not interchangeable the criteria listed in MIL-D-70327 shall apply and new drawings shall be prepared as required.

3.5 PERFORMANCE. The adapter shall be capable of adapting the Analyzer, Test Set TS-1304/GWM-4 of the Test Set, Armament Wiring System, Special Weapon AN/GWM-4 to the Aircraft Wiring System F-100 and F-105/GAM-83A or F-100 and F-105/GAM-83B.

3.6 ELECTRICAL REQUIREMENTS

3.6.1 ELECTRICAL CONTINUITY. Electrical continuity shall be in each circuit.

3.6.2 INSULATION RESISTANCE. There shall be no less than 20 megohms measured resistance between individual pins and shell of their respective connectors or between individual pins and all other pins within a connector.

3.7 ENVIRONMENTAL REQUIREMENTS. The adapter shall be capable of satisfactory performance, during or after, as applicable, subjection to the following environmental conditions:

a. High temperature up to 71°C (160°F) for a period of 50 hours.

b. Low temperature between -54°C (-65°F) and -62°C (-80°F) for a period of 72 hours.

c. Humidity of 95 percent with temperatures ranging from 20°C (68°F) to 71°C (160°F).

d. Rain as encountered in service use.

e. Sand and dust as encountered under service conditions in desert areas.

f. Salt spray comparable to salt sea atmosphere.

g. Altitude up to 6000 feet above sea level.

h. Vibration such as that encountered in transportation and service use.

i. Shock equal to being dropped from a vehicle when in transport configuration.

j. Operation in explosive mixtures in surrounding atmosphere without initiating an explosion.

k. Fungus as found in tropical or damp climate.

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3.8 IDENTIFICATION OF PRODUCT. Equipment assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.9 REFERENCE DESIGNATIONS. Reference designations used for markings within the unit shall be in accordance with MIL-STD-16. Markings shall not adversely affect the leakage path between conductors, or any other factors of performance. Reference symbols shall be employed to identify each part for its particular circuit application. The same reference symbol shall be used for identifying a particular part on all types of data, including drawings, diagrams, and spare parts prepared for the equipment. When practicable, markings shall be placed on the supporting structure adjacent to each replaceable part in such a position as to be readily viewed.

3.10 WORKMANSHIP. The adapter, including all parts and accessories, shall be fabricated and finished in a manner that will insure that the criteria of appearance, fit, and adherence to specified tolerances are observed. These criteria must apply to all accessories and to all external non-functioning parts; e.g., cases, covers, and nameplates, as well as to functional parts. Particular attention shall be given to the neatness and thoroughness of marking of parts and assemblies, plating, painting, machine screw assemblage, soldering, wiring, and freedom of parts from burrs and sharp edges which may mar associated equipment or cause injury to operating personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. The inspection and testing of the adapter shall be classified as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test samples shall consist of models representative of the production adapters. They shall be tested at a laboratory designated by the procuring activity, or when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.1.1 PREPRODUCTION TEST REPORT. In the event that preproduction testing is conducted at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

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4.2.2 PREPRODUCTION TESTS. Preproduction tests shall consist of all tests described in "4.5 TEST METHODS" of this specification.

4.3 ACCEPTANCE TESTS. Acceptance tests shall consist of individual tests only.

4.3.1 INDIVIDUAL TESTS. Each adapter shall be subjected to the following:

a. Examination of Product

b. Conformance to Drawings

c. Continuity Test

d. Insulation Resistance

4.3.2 DEFECTS IN ITEMS ALREADY ACCEPTED. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 TEST CONDITIONS

4.4.1 ATMOSPHERIC CONDITIONS. Unless otherwise specified in specific tests, all tests required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25°C (77°F) ± 1°C (2°F). Relative humidity shall be 80 percent or less. Allowance shall be made for deviation from the foregoing conditions if such deviations exist.

4.4.2 TEST POWER. Unless otherwise specified in specific tests, the testing power applied shall be 28 v ± 1v dc.

4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT. Each adapter shall be examined to determine compliance with the requirements of this specification including materials, workmanship and ease of operation of controls.

4.5.2 CONFORMANCE TO DRAWINGS. Each adapter shall be inspected to determine strict conformance to the engineering drawings and specifications and standards referenced thereon.

4.5.3 CONTINUITY TEST. Continuity of each circuit shall be checked by means of any circuit checking device with an impressed voltage of

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6.0 volts or less. Circuitry of the adapter is shown in drawing 60D46021. Indication of "No Continuity" in any completed circuit shall be cause for rejection.

4.5.4 INSULATION RESISTANCE TEST. Insulation resistance of the adapter shall be checked in accordance with method 302, condition B, MIL-STD-202. Duration of electrification shall be one minute. Measured resistance of less than 20 megohms shall be cause for rejection.

4.5.5 ENVIRONMENTAL TESTS, ADAPTER. The adapter, less cables, shall be subjected to the environmental tests listed below. When the test requires operation of equipment, operation shall consist of the continuity and insulation resistance tests of specification unless otherwise specified herein for a specific test. Failure to satisfy the continuity and insulation resistance tests shall be considered as failure to satisfy the environmental test and shall be cause for rejection.

TEST	SPECIFICATION	PROCEDURE
High Temperature	MIL-E-4970	I
Low Temperature	MIL-E-4970	I
Humidity	MIL-E-4970	I
Rain	MIL-E-4970	I
Sand and Dust	MIL-E-4970	I
Salt Fog	MIL-E-4970	
Vibration	MIL-E-4970	VI
Exception: Lower frequency shall be 5 cps.		
Shock	MIL-T-4807	Method 3A
Explosion Proof	MIL-E-4970	III
Fungus Resistance	MIL-E-4970	I

NOTE: The adapter selected for this test shall not have been subjected to the Sand and Dust or Salt Fog Tests.

4.5.5.1 CABLES. The cables shall satisfy all tests described in MIL-C-25200 prior to installation.

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5. PREPARATION FOR DELIVERY

5.1 PRESERVATION, PACKAGING AND PACKING. Packing and packaging for shipment shall be in accordance with MIL-E-17555. Levels of protection shall be as specified by the procuring activity.

5.2 MARKING FOR SHIPMENT AND STORAGE. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be:

ADAPTER, TEST MX-3413/GWM-4

6. NOTES

6.1 INTENDED USE. The Adapter, Test MX-3413/GWM-4 with appropriate combinations of cables is intended to adapt the Analyzer TS-1304/GWM-4 to the F-100 and F-105/GAM-83A and F-100 and F-105/GAM-83B Special Weapon Monitor and Control Aircraft Wiring System.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests will be performed by the procuring activity or by the contractor under the supervision of the procuring activity.

c. Levels of preservation, packaging and packing required.

6.2.1 PREPRODUCTION TEST SAMPLES. It is expected that the contract or purchase order will specify that three units representative of the production adapter will be submitted to the procuring activity for test and evaluation. The invitation for bids and the contract or purchase order should specify the point of inspection for these tests.

6.3 PRODUCTION DELIVERY. No production items should be delivered prior to approval of the preproduction samples.

6.4 SOURCES. When requesting publications, refer to both title and number.

6.5 DEFINITIONS

a. PREPRODUCTION TESTS. Preproduction tests are those tests performed prior to production on a representative production adapter to demonstrate that the equipment meets the requirements of this specification.

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b. ACCEPTANCE TESTS. Acceptance tests are those test accomplished on items of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the units are equivalent in performance and construction to the approved preproduction items.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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